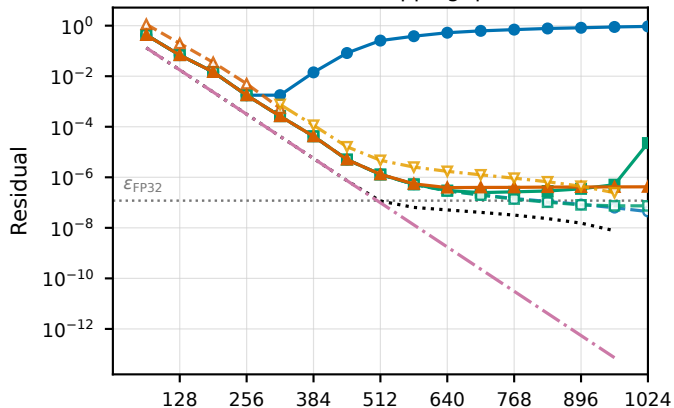


- BCGS Range Finder: $\|A - QQ^T A\|_2$ (actual)
- BCGS Range Finder: $\|A - QB\|_F$ (tracked)
- BCGS Range Finder + Reorth.: $\|A - QQ^T A\|_2$ (actual)
- BCGS Range Finder + Reorth.: $\|A - QB\|_F$ (tracked)
- HH Range Finder: $\|A - QB\|_2$ (actual)

- HH Range Finder: $(\|A\|_F^2 - \sum_i \|B_i\|_F^2)^{\frac{1}{2}}$ (tracked)
- HH Range Finder: $\|B_{i+1:j}\|_F$ (tracked)
- Actual $\sigma_{k+1}(A)$
- Prescribed $\sigma_{k+1}(A)$

- BCGS Range Finder
- BCGS Range Finder + Reorth.
- HH Range Finder

(a) Residual and stopping quantities



Number of acquired basis vectors k

(b) Loss of orthogonality

